IRON WITHIN RESEARCH

KPV — Certificates of Analysis

Every batch we have tested for this product, newest first.
Match the strength on your vial to the rows below and turn to that page.

PAGE	STRENGTH	BATCH / LOT
2–3	21mg, 30mg	10/9/2026 · 30mg · Lot IWR-1010-KPV3
4–5	20mg, 48mg	10/9/2026 · 20mg · Lot IWR-1010-KPV2
6–7	10mg, 53mg	10/9/2026 · 10mg · Lot IWR-1010-KPV1
8–9	20mg, 22mg	9/13/2026 · 20mg · Lot IWR-090926KPV3
10–11	10mg, 45mg	9/13/2026 · 10mg · Lot IWR-090926KPV2
12–13	10mg, 87mg	9/13/2026 · 10mg · Lot IWR-090926KPV1
14–15	10mg, 78mg	8/13/2026 · Lot IWR-081026KPV
16–20	10mg, 15mg, 58mg	8/2/2026 · Lot IWR-2607-KPV10L
21–22	5mg, 10mg, 11mg, 18mg	5/6/2026

Page numbers include this index page. Testing is per production batch; one batch can cover several vial sizes.

SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0298

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
KPV — 30 mg

PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-1010-KPV3	99.65%	30.21 mg	Confirmed

COA #:	SBA-26-0298	Method:	Full QC Panel
Lot Number:	IWR-1010-KPV3	Analysis Date:	2026-10-09
Labeled Content:	30 mg	Date Received:	2026-10-06
		Appearance:	White lyophilized powder, intact seal

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

Test	Specification	Result
Identity	KPV	Confirmed
Purity (HPLC)	>= 98.0%	99.65%
Content (per vial)	Report only	30.21 mg/vial
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.03 EU/mL
Sterility Screen (rapid PCR)	No growth	No growth
Heavy Metals (ICP-MS)	ICH Q3D limits	All analytes within limits



SAMPLE AS RECEIVED

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.

Kasey Malganius
Lab Director
2026-10-09

COA #:	**SBA-26-0298**
Access Code:	**SMMVTRLZ**
Analyst:	**A. Joseph**
Verify:	summitbioanalytics.com/verify
Issued:	2026-10-09

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0305

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
KPV — 20 mg

PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-1010-KPV2	99.81%	20.48 mg	Confirmed

COA #:	SBA-26-0305	Method:	Full QC Panel
Lot Number:	IWR-1010-KPV2	Analysis Date:	2026-10-09
Labeled Content:	20 mg	Date Received:	2026-10-06
		Appearance:	White lyophilized powder, intact seal

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

Test	Specification	Result
Identity	KPV	Confirmed
Purity (HPLC)	>= 98.0%	99.81%
Content (per vial)	Report only	20.48 mg/vial
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.03 EU/mL
Sterility Screen (rapid PCR)	No growth	No growth
Heavy Metals (ICP-MS)	ICH Q3D limits	All analytes within limits



SAMPLE AS RECEIVED

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.

Kasey Malganius

Kasey Malganius
Lab Director
2026-10-09

COA #:	**SBA-26-0305**
Access Code:	**MHSYB9FN**
Analyst:	**A. Joseph**
Verify:	summitbioanalytics.com/verify
Issued:	2026-10-09

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SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0301

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TESTED FOR
Iron Within Research

SAMPLE
KPV — 10 mg

PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-1010-KPV1	99.45%	10.53 mg	Confirmed

COA #:	SBA-26-0301	Method:	Full QC Panel
Lot Number:	IWR-1010-KPV1	Analysis Date:	2026-10-09
Labeled Content:	10 mg	Date Received:	2026-10-06
		Appearance:	White lyophilized powder, intact seal

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

Test	Specification	Result
Identity	KPV	Confirmed
Purity (HPLC)	>= 98.0%	99.45%
Content (per vial)	Report only	10.53 mg/vial
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.04 EU/mL
Sterility Screen (rapid PCR)	No growth	No growth
Heavy Metals (ICP-MS)	ICH Q3D limits	All analytes within limits



SAMPLE AS RECEIVED

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.

Kasey Malganius

Kasey Malganius
Lab Director
2026-10-09

COA #:	**SBA-26-0301**
Access Code:	**KQGQ43GR**
Analyst:	**A. Joseph**
Verify:	summitbioanalytics.com/verify
Issued:	2026-10-09

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0183

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
KPV — 20 mg

PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-090926KPV3	99.97%	21.22 mg	Confirmed

COA #:	SBA-26-0183	Method:	Full QC Panel
Lot Number:	IWR-090926KPV3	Analysis Date:	2026-09-13
Labeled Content:	20 mg	Date Received:	2026-09-07
		Appearance:	White lyophilized powder, intact seal
		Vial Cap:	Purple #760A7D

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

Test	Specification	Result
Identity	KPV	Confirmed
Purity (HPLC)	>= 98.0%	99.97%
Content (per vial)	Report only	21.22 mg/vial
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.03 EU/mL
Sterility Screen (rapid PCR)	No growth	No growth
Heavy Metals (ICP-MS)	ICH Q3D limits	All analytes within limits

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.

Kasey Malganius

Kasey Malganius
Lab Director
2026-09-13

COA #:	**SBA-26-0183**
Access Code:	**AA2CHW25**
Analyst:	**R. Jeffereson**
Verify:	summitbioanalytics.com/verify
Issued:	2026-09-13

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0182

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
KPV — 10 mg

PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-090926KPV2	99.93%	10.45 mg	Confirmed

COA #:	SBA-26-0182	Method:	Full QC Panel
Lot Number:	IWR-090926KPV2	Analysis Date:	2026-09-13
Labeled Content:	10 mg	Date Received:	2026-09-07
		Appearance:	White lyophilized powder, intact seal
		Vial Cap:	Black #010101

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

Test	Specification	Result
Identity	KPV	Confirmed
Purity (HPLC)	>= 98.0%	99.93%
Content (per vial)	Report only	10.45 mg/vial
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.02 EU/mL
Sterility Screen (rapid PCR)	No growth	No growth
Heavy Metals (ICP-MS)	ICH Q3D limits	All analytes within limits

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.

Kasey Malganius

Kasey Malganius
Lab Director
2026-09-13

COA #:	**SBA-26-0182**
Access Code:	**RWWZ4U83**
Analyst:	**R. Jeffereson**
Verify:	summitbioanalytics.com/verify
Issued:	2026-09-13

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SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0181

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
KPV — 10 mg

PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-090926KPV1	99.32%	10.87 mg	Confirmed

COA #:	SBA-26-0181	Method:	Full QC Panel
Lot Number:	IWR-090926KPV1	Analysis Date:	2026-09-13
Labeled Content:	10 mg	Date Received:	2026-09-07
		Appearance:	White lyophilized powder, intact seal
		Vial Cap:	Yellow #E0C04A

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

Test	Specification	Result
Identity	KPV	Confirmed
Purity (HPLC)	>= 98.0%	99.32%
Content (per vial)	Report only	10.87 mg/vial
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.03 EU/mL
Sterility Screen (rapid PCR)	No growth	No growth
Heavy Metals (ICP-MS)	ICH Q3D limits	All analytes within limits

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.

Kasey Malganius

Kasey Malganius
Lab Director
2026-09-13

COA #:	**SBA-26-0181**
Access Code:	**F8EJJEAE**
Analyst:	**R. Jeffereson**
Verify:	summitbioanalytics.com/verify
Issued:	2026-09-13

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SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0085

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
KPV — 10 mg

PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-081026KPV	99.87%	10.78 mg	Confirmed

COA #:	SBA-26-0085	Method:	Full QC Panel
Lot Number:	IWR-081026KPV	Analysis Date:	2026-08-13
Labeled Content:	10 mg	Date Received:	2026-08-10
		Appearance:	White lyophilized powder, intact seal
		Vial Cap:	Purple #760A7D

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

Test	Specification	Result
Identity	KPV	Confirmed
Purity (HPLC)	>= 98.0%	99.87%
Content (per vial)	Report only	10.78 mg/vial
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.04 EU/mL
Sterility Screen (rapid PCR)	No growth	No growth
Heavy Metals (ICP-MS)	ICH Q3D limits	All analytes within limits

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.

Kasey Malganius

Kasey Malganius
Lab Director
2026-08-13

COA #:	**SBA-26-0085**
Access Code:	**94P86JF4**
Analyst:	**A. Grinder**
Verify:	summitbioanalytics.com/verify
Issued:	2026-08-13

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

Horizon Analytical

Certificate of Analysis

Lot Number: IWR-4451178-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

Received Date: 06/29/2026
Analysis Conducted: 06/24/2026
Searchable via: horizonanalytical.com

Compound:	KPV
Lot:	IWR-4451178-P
Appearance:	White Lyophilized Powder

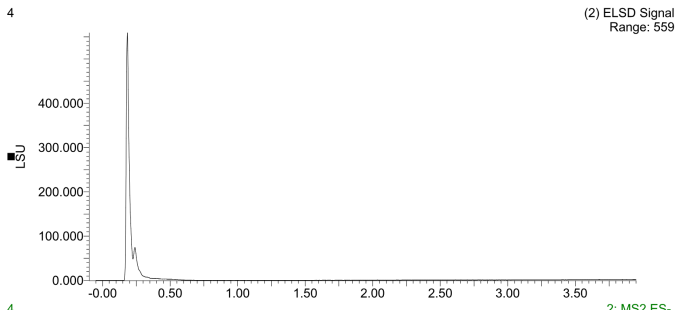
CAS:	112965-21-6
Formula:	$C_{17}H_{32}N_6O_4$
Mol Weight:	~384.48 g/mol

Pubchem CID: 9929972

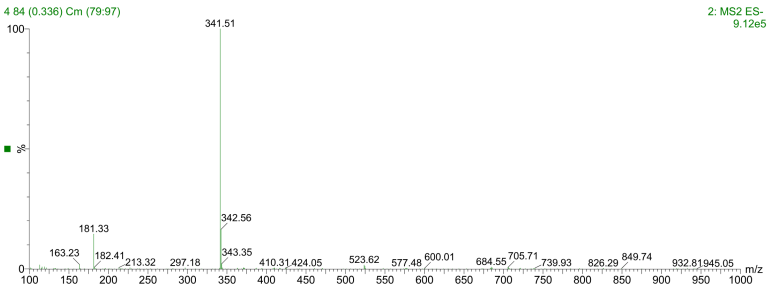
Qualitative and Quantitative chemical analysis by Ultra High Performance Liquid Chromatography with Mass Spectrometry

	Specification	Result	Scan to Validate:
Compound Test:	KPV	KPV	
Quantity:	10mg	10.15mg	
Purity:	>98%	99.28%	

UPLC



MS



Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

This purity analysis was conducted using UPLC/MS under standard laboratory conditions, following validated analytical protocols to ensure accurate and reliable results. This analysis is intended for informational and research applications.

Horizon Analytical

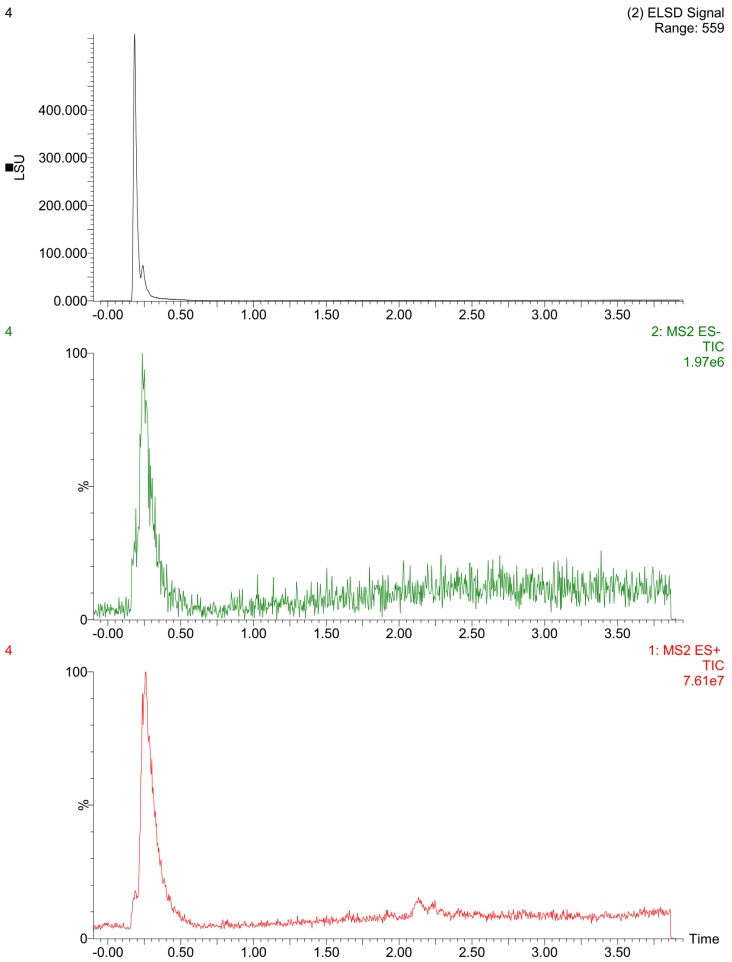
Certificate of Analysis

Lot Number: IWR-4451178-P
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

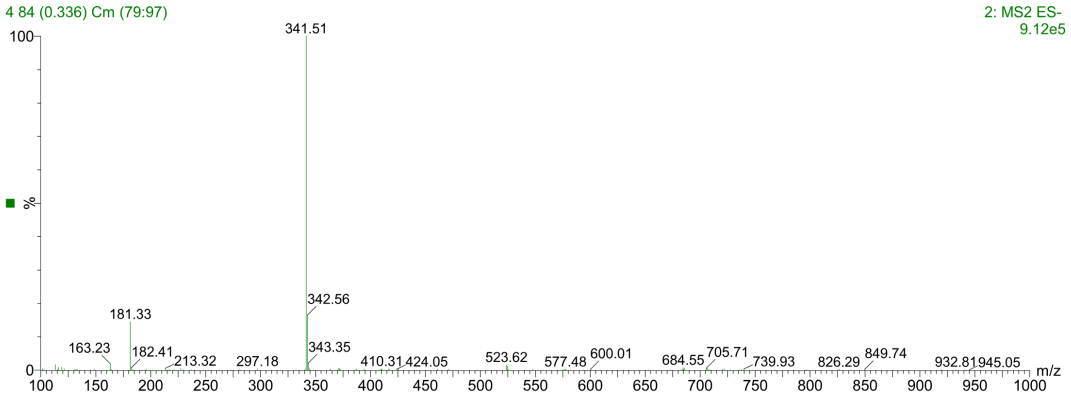
Received Date: 06/29/2026
Analysis Conducted: 06/24/2026
Searchable via: horizonanalytical.com

KPV (10mg) • Pubchem CID: 9929972

Ultra High Performance Liquid Chromatography (UPLC)



Mass Spectrometry (MS)



Horizon Analytical

Certificate of Analysis

Lot Number: IWR-4451178-E
Client Name: Iron Within Research
Identity: www.ironwithinresearch.com

Received Date: 06/29/2026
Analysis Conducted: 06/24/2026
Searchable via: horizonanalytical.com

Compound:	KPV
Lot:	IWR-4451178-E
Appearance:	-

CAS:	112965-21-6
Formula:	$C_{17}H_{32}N_6O_4$
Mol Weight:	~384.48 g/mol

Pubchem CID: 9929972

Endotoxin Test

	Specification	Result	Scan to Validate:
Compound Test:	KPV	-	
Endotoxin:	-	0.075 EU/mL	

Aleksey Yevtodiyenko PhD
Research and Formulation Chemist

This endotoxin analysis was performed under standard laboratory conditions using validated testing methodologies to ensure accurate and reliable results. The analysis is intended for informational and research purposes only.

Contact at: contact@horizonanalytical.com

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SUMMIT
BIOANALYTICS

Certificate of Analysis

#SBA-26-0018

Summit BioAnalytics | Nassau, The Bahamas | lab@summitbioanalytics.com | summitbioanalytics.com

TESTED FOR
Iron Within Research

SAMPLE
KPV — 10 mg

PASS

BATCH #	PURITY	MASS	IDENTITY
IWR-2607-KPV10	99.04%	10.58 mg	Confirmed

COA #:	SBA-26-0018	Method:	Full QC Panel
Lot Number:	IWR-2607-KPV10	Analysis Date:	2026-08-02
Labeled Content:	10 mg	Date Received:	2026-07-29
		Appearance:	White lyophilized powder, intact seal

SCAN TO VERIFY

- ✓ Identity
- ✓ Purity
- ✓ Endotoxins
- ✓ Fentanyl Free

VIAL CAP
Pink
#FB84B0

CLOSURE AS RECEIVED

Test	Specification	Result
Identity	KPV	Confirmed
Purity (HPLC)	>= 98.0%	99.04%
Content (gross)	Report only	10.58 mg
Fentanyl Screen (LC-MS/MS)	Not detected	Not detected
Endotoxins (LAL)	Report Result	0.09 EU/mL

Test	Specification	Result
Sterility Screen (rapid PCR)	No growth	**No growth**
Heavy Metals (ICP-MS)	ICH Q3D limits	**All analytes within limits**

Qualitative and quantitative chemical analysis by RP-HPLC with UV detection; identity confirmed by LC-MS. Fentanyl screening by targeted LC-MS/MS. Sterility by rapid nucleic-acid amplification — a screening method, not compendial sterility per USP <71>. Full methodology available on request.



Kasey Malganius
Lab Director
2026-08-02

COA #:	**SBA-26-0018**
Access Code:	**J52KQLQU**
Analyst:	**R. Whitfield**
Verify:	summitbioanalytics.com/verify
Issued:	2026-08-02

This certificate relates only to the item tested, as received. Research use only — not for human or veterinary use. A not-detected screening result describes this sample only and is not a statement about any other unit or lot.

FREEDOM DIAGNOSTICS

Certificate of Analysis

FreedomDiagnosticsTesting.com

Admin@FreedomDiagnostics.net



Client:	Iron Within Research
Accession #:	2605060164
Search Code:	Iron2605060164
Received:	05/06/2026
Reported:	05/08/2026
Lot:	Grey Cap

Sample Summary

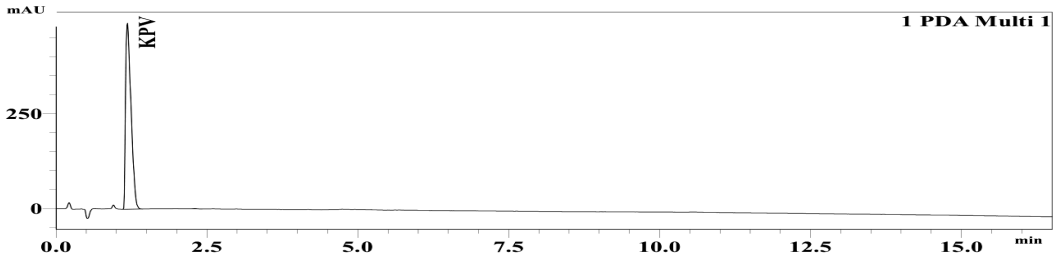
Product:	KPV 10mg	**Purity:**	99.89%
Identity:	Confirmed	**Net Content:**	13.18 mg
Appearance:	White Lyophilized Powder		

Analytical Results

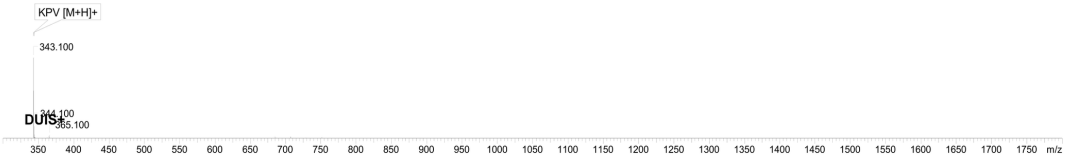
Test	Result
Identity (LC-MS)	KPV
Purity (HPLC-UV)	99.89%
Net Content	13.18 mg

Method: HPLC with UV detection coupled with mass spectrometry (LC-MS).

Chromatogram



Mass Confirmation



Alex Johnson

Principal Chemist

FreedomDiagnosticsTesting.com

Admin@FreedomDiagnostics.net

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The peptide purity analysis reported here was conducted using LCMS/MS under standard laboratory conditions. This analysis is intended for informational purposes only and is specific to the sample(s) provided. The peptides tested are intended for research use only and are not approved for human or veterinary use, diagnostic, therapeutic, or clinical applications. Results should be interpreted by qualified professionals within the scope of the intended research. The accuracy and reliability of the test may be influenced by sample integrity, handling, and other experimental variables.

FREEDOM DIAGNOSTICS

Certificate of Analysis

FreedomDiagnosticsTesting.com

Admin@FreedomDiagnostics.net



Client:	Iron Within Research
Accession #:	2605060163
Search Code:	Iron2605060163
Received:	05/06/2026
Reported:	05/08/2026
Lot:	Yellow Cap

Sample Summary

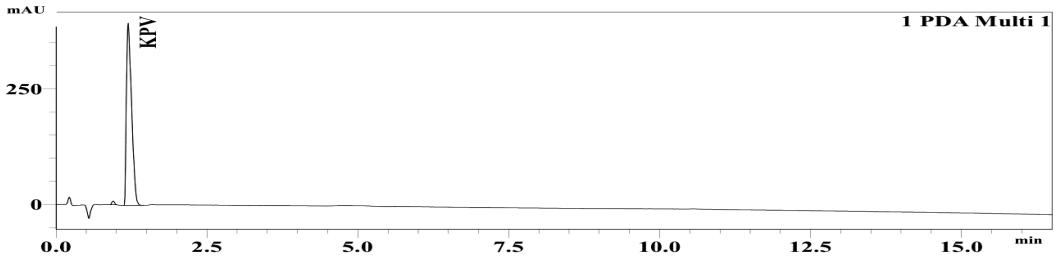
Product:	KPV 5mg	**Purity:**	99.01%
Identity:	Confirmed	**Net Content:**	4.11 mg
Appearance:	White Lyophilized Powder		

Analytical Results

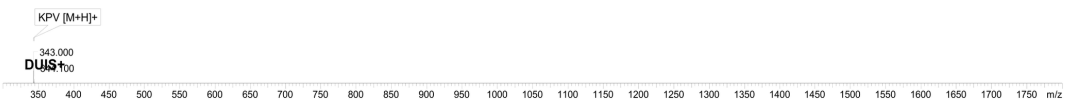
Test	Result
Identity (LC-MS)	KPV
Purity (HPLC-UV)	99.01%
Net Content	4.11 mg

Method: HPLC with UV detection coupled with mass spectrometry (LC-MS).

Chromatogram



Mass Confirmation



Alex Johnson

Principal Chemist

FreedomDiagnosticsTesting.com

Admin@FreedomDiagnostics.net

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